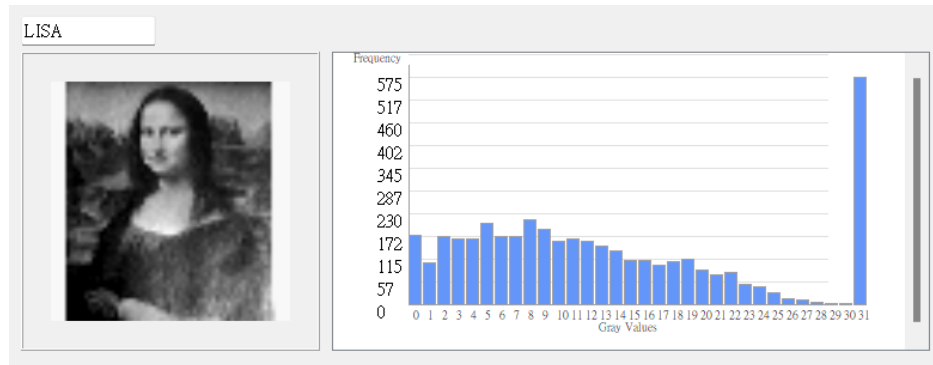


Digital Image Processing

Homework 1

PART1: Histogram of an Image

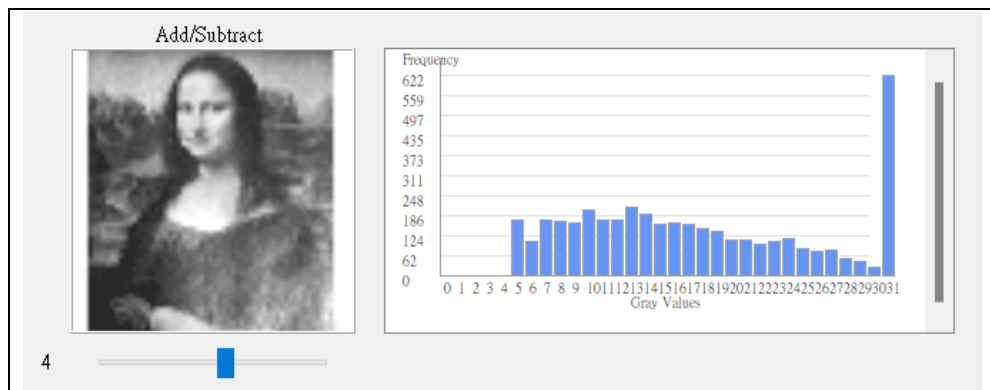
1. 讀取.64 文件並將文件中 0~9 A~V 的 ASCII 轉換為 0~31 的灰階值，使用生成的 64*64 二維陣列繪製圖像
2. 統計 32 個灰階值出現的頻率，使用 Qt 作直方圖

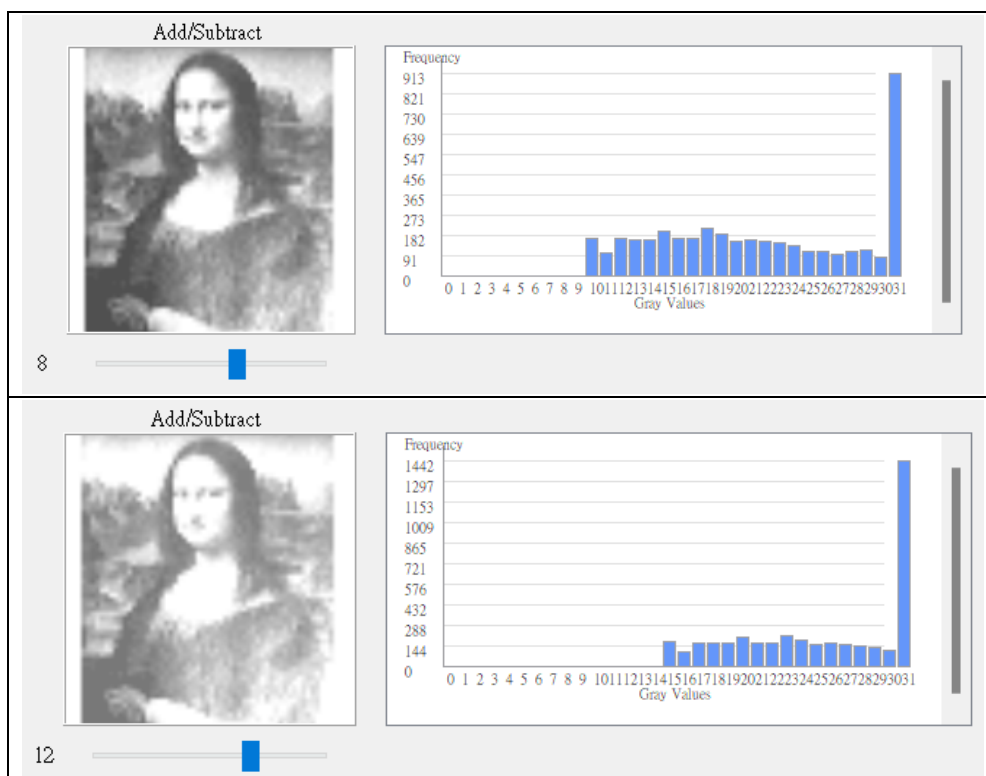


PART2: Arithmetic Operations of an Image Array

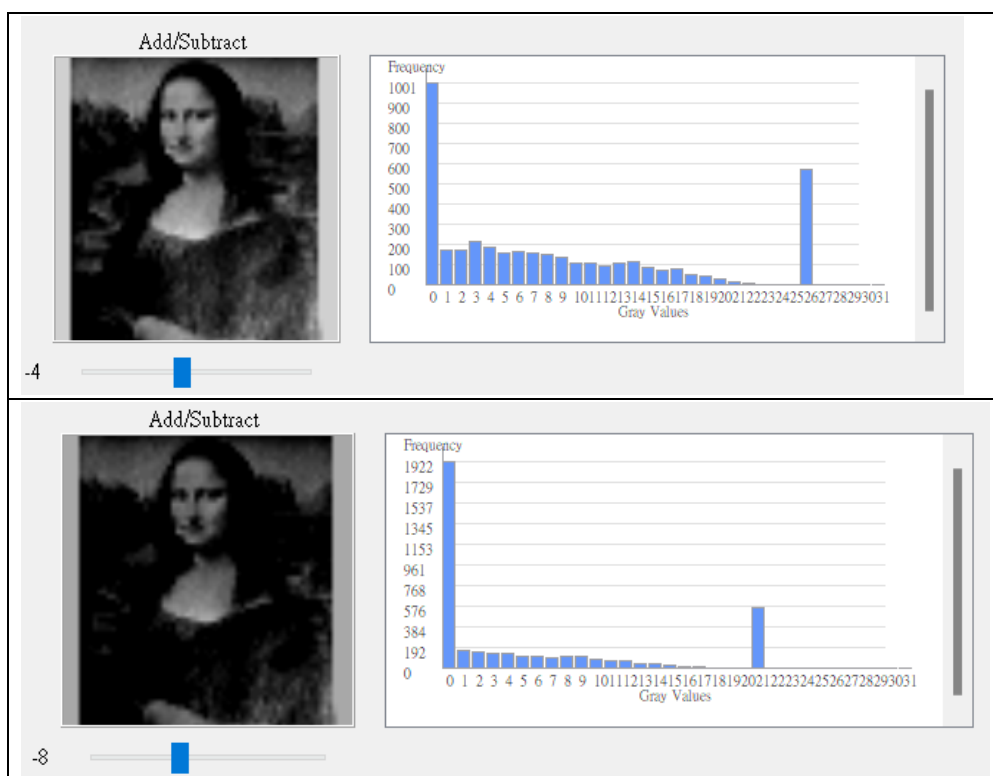
1. Add or subtract a constant value to each pixel in the image

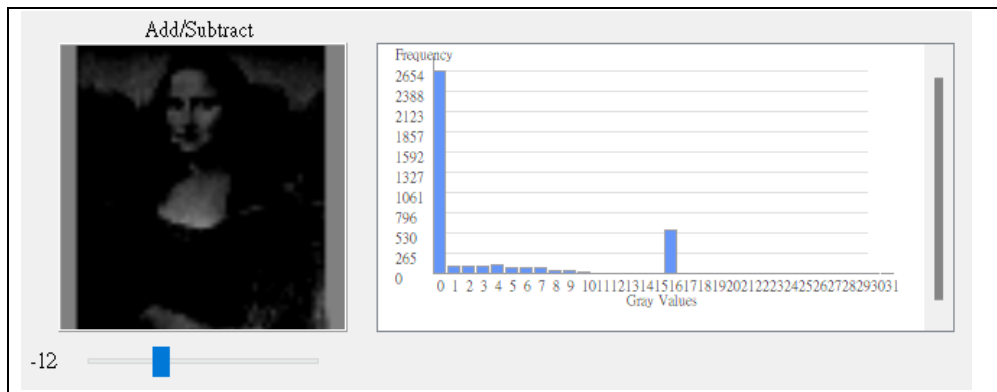
將原本的所有灰階值加上一個固定常數，且最大值固定為 31，從直方圖可見所有數值右移，最終逐漸變為全白圖像





將原本的所有灰階值減上一個固定常數，且最小值固定為 0，從直方圖可見所有數值左移，最終逐漸變為全黑圖像

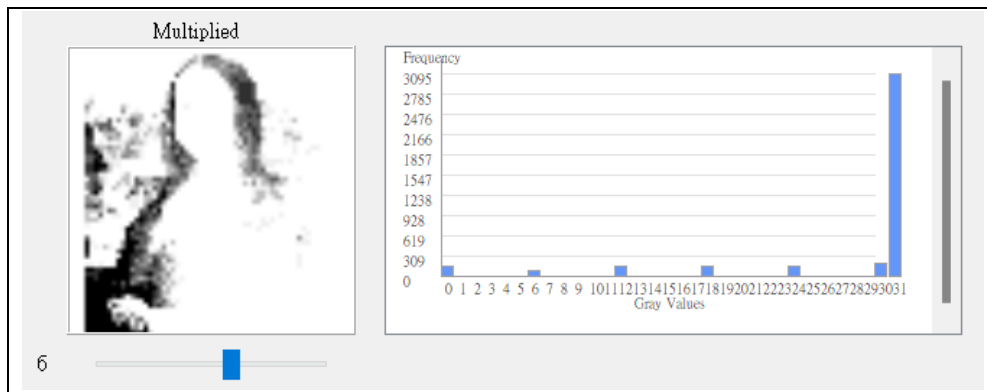




2. Multiply a constant to each pixel in the image.

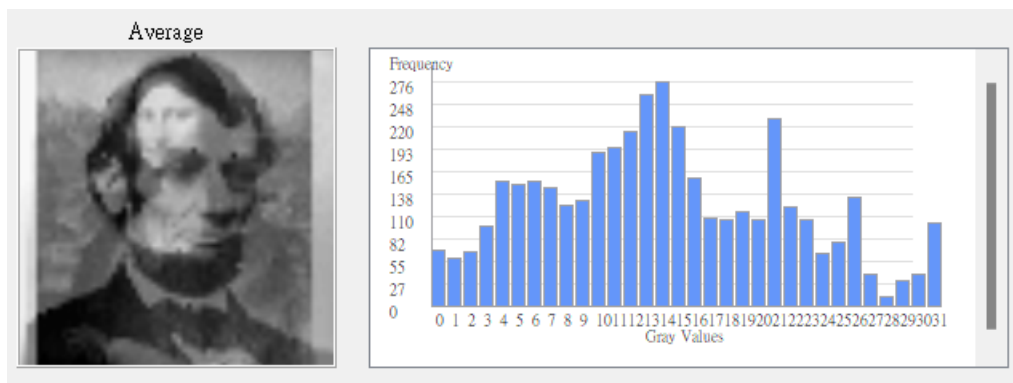
將原本的所有灰階值乘上一個固定常數，從直方圖可見所有變得分散，最終逐漸圖像整體變白，但存在原為 0 的數值，經過乘法後仍為 0





3. Create a new image which is the average image of two input images.

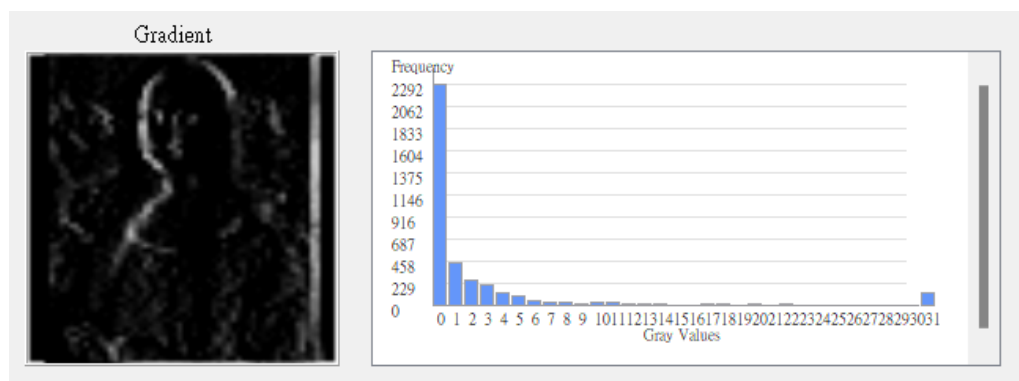
將兩張影像相加後取得平均，兩張圖像變成半透明，得出兩者的平均直方圖



4. Create a new image $g(x,y)$ in which the value of each pixel is determined by calculating the pixel values of the input image $f(x,y)$ using the following

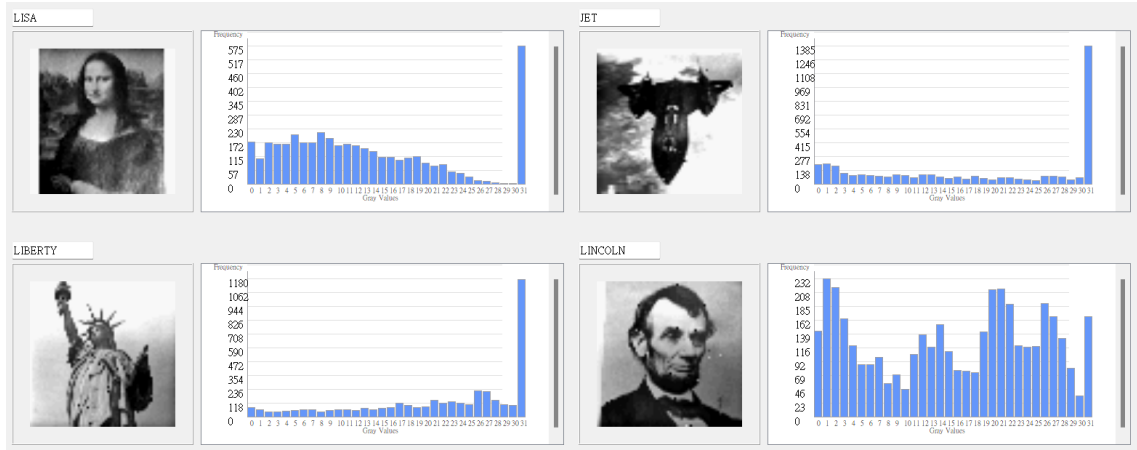
$$\text{equation: } g(x,y) = f(x,y) - f(x-1,y)$$

根據指定函數，將灰階值減去前一行的灰階值作為左方的新數值，此方法會擴大數值差異，凸顯影像中物體的邊界



Qt UI

PART1:



PART2:

